Nishal Stanislaus Silva, *PhD*

Machine Learning Scientist | Applied ML, Deep Learning, Time-Series | Peer-Reviewed Research

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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19th October 2026

Re: Machine Learning Scientist, Advanced Technology – Alberta Machine Intelligence Institute (Amii)

Dear Hiring Team,

I am applying for the Machine Learning Scientist, Advanced Technology role at Amii. I am a machine learning scientist with a PhD (University of Trento, 2025), 10+ years across industry and academia, and 10+ peer-reviewed publications. Your posting pairs client-facing applied ML with research and manuscripts, which is how I have worked. At MAS Holdings I delivered ML and computer vision systems to operations and engineering stakeholders, and at Trento I moved research prototypes into documented, deployable components with industry partners.

The accomplishment I am most proud of is the real-time audio pattern detection system published in the Journal of the Audio Engineering Society (2026). It runs at 14 ms on a Raspberry Pi 4 with F1 0.76, 74x faster than the DTW baseline. I am proud of it because it went the whole way: from a research question, through benchmark suites with baselines and ablations, to a system performers used in two live concerts, and then to four open datasets on Zenodo that other researchers can build on.

Several requirements in your posting match my work directly. For model development, I work in PyTorch, scikit-learn, Pandas and SciPy, and I have deployed on AWS (SageMaker, ECS, MWAA). For client work, I turned compliance and operations requirements into technical specifications at Forestpin and MAS, and I explained results to non-technical stakeholders. For mentoring, I mentored engineers at MAS, supervised two BSc theses at Trento, and I review for IEEE Access and sit on programme committees for the IEEE Internet of Sounds Symposium and the Immersive and 3D Audio conference. My human-centred evaluation work on usability and interpretability is the closest experience I have to your Principled AI Framework.

I should be direct about the gaps. I have not worked professionally with reinforcement learning, and I have not used Optuna, Weights & Biases, LightGBM or XGBoost. Most of my modelling has been supervised sequence models, with prototypes in VAE and diffusion methods for limited-label data. I expect to pick up reinforcement learning quickly with a team of experts around me, and I would be glad to start on client problems where my evaluation and deployment experience applies right away.

I am a Canadian permanent resident, eligible to work without sponsorship, available immediately, and open to relocating to Edmonton. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD